Output-error sensitivity of a Clements unitary mesh vs. phase-shifter error (small range)

$N = 20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500$ | Haar-random unitary U (single Clements mesh) | one forward pass $y = Ux$ | $\hat{y}$ = complex output field, normalised to unit L2 norm | $E = \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2$

Inputs per unitary: 7 types x 10 random draws (dense complex, dense real, binary ±1, phase-only, sparse 10%, single port, Gaussian beam) | shown: mean over all 7 types (equal weight)

30 hardware realizations per unitary per point (each chip sees one input of every type; each input used on 3 chips) | 25 Haar unitaries | 50 points in $\sigma_{\mathrm{rad}} \in [0, 0.25]$ rad | all other error channels ideal (zero)

Hardware model: i.i.d. Gaussian phase errors $\delta\theta$, $\delta\phi \sim \mathcal{N}(0, \sigma_{\mathrm{rad}}^2)$ on every MZI, independent for every chip; single Clements mesh, ideal 50:50 couplers, no loss, ideal output phase screen

Each point: $E_{\mathrm{rms}} = \sqrt{\langle E^2 \rangle}$ over unitaries, chips and input types; band: ±1 s.d. across Haar-random targets (s.e. in the CSV)

